Logistic Regression and Classification

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Logistic Regression & Classification Practice Exercises**

In evaluating a binary classifier, the confusion matrix shows $TP = 60$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Final Answer: 83.3%. Accuracy is the ratio of correct predictions to total samples:

$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{60 + 40}{120} = \frac{100}{120} = 83.3\%.$

2 **[Foundational]** • **Target: The Odds Ratio and Logit Formulation**

In evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Final Answer: 84.6%. Accuracy is the ratio of correct predictions to total samples:

$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{70 + 40}{130} = \frac{110}{130} = 84.6\%.$

3 **[Foundational]** • **Target: Binary Cross-Entropy Loss**

In evaluating a binary classifier, the confusion matrix shows $TP = 45$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Final Answer: 81.0%. Accuracy is the ratio of correct predictions to total samples:

$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{45 + 40}{105} = \frac{85}{105} = 81.0\%.$

4 **[Foundational]** • **Target: The Confusion Matrix and Core Metrics**

In evaluating a binary classifier, the confusion matrix shows $TP = 55$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Final Answer: 82.6%. Accuracy is the ratio of correct predictions to total samples:

$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{55 + 40}{115} = \frac{95}{115} = 82.6\%.$

5 **[Foundational]** • **Target: The F_1-Score, Thresholding and ROC-AUC**

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.30$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

Final Answer: -0.847. Odds are $\frac{p}{1-p} = \frac{0.30}{0.70} = 0.4286$. Taking natural logarithm: $\ln(0.4286) = -0.847$.

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6 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.40$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

Final Answer: -0.405. Odds are $\frac{p}{1-p} = \frac{0.40}{0.60} = 0.6667$. Taking natural logarithm: $\ln(0.6667) = -0.405$.

7 [Foundational] • Target: The Odds Ratio and Logit Formulation

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.50$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

Final Answer: 0.000. Odds are $\frac{p}{1-p} = \frac{0.50}{0.50} = 1.0000$. Taking natural logarithm: $\ln(1.0000) = 0.000$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Binary Cross-Entropy Loss

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.60$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

Final Answer: 0.405. Odds are $\frac{p}{1-p} = \frac{0.60}{0.40} = 1.5000$. Taking natural logarithm: $\ln(1.5000) = 0.405$.

9 [Intermediate] • Target: The Confusion Matrix and Core Metrics

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.70$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

Final Answer: 0.847. Odds are $\frac{p}{1-p} = \frac{0.70}{0.30} = 2.3333$. Taking natural logarithm: $\ln(2.3333) = 0.847$.

10 [Intermediate] • Target: The F_1-Score, Thresholding and ROC-AUC

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.80$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

Final Answer: 1.386. Odds are $\frac{p}{1-p} = \frac{0.80}{0.20} = 4.0000$. Taking natural logarithm: $\ln(4.0000) = 1.386$.

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11 **[Intermediate]** • **Target: Logistic Regression & Classification Practice Exercises**

Consider a binary classification problem where the decision boundary is defined by the logistic function $p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$, which represents the probability of a data point $\mathbf{x} = (x_1, x_2)$ belonging to class 1. The estimated model parameters are $\beta_0 = -3$, $\beta_1 = 1$, and $\beta_2 = -2$. We are interested in the geometric shape of the classification boundary. The classification boundary occurs where $p(x) = 0.5$. Find the equation of this boundary in the x_1x_2-plane.

Final Answer: $x_2 = \frac{1}{2}x_1 - \frac{3}{2}$. The logistic regression model gives the probability of belonging to class 1 as $p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$. The classification boundary is the set of points where $p(x) = 0.5$. Setting $p(x) = 0.5$, we have: $0.5 = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$ This implies: $1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)} = 2$ Therefore: $e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)} = 1$ Taking the natural logarithm.

12 **[Intermediate]** • **Target: The Odds Ratio and Logit Formulation**

A binary classification model uses a logistic sigmoid function $\sigma(z) = \frac{1}{1 + e^{-z}}$. The decision boundary is defined by $\sigma(z) = 0.5$, where $z = w_1x_1 + w_0$. For a dataset with two features x_1 and x_2, the model's linear component is $z = 2x_1 - x_2 + b$. If the classification boundary passes through the point $(x_1, x_2) = (3, 1)$, what is the value of the bias b? Show your complete algebraic work.

Final Answer: $b = -7$. The logistic sigmoid function is $\sigma(z) = \frac{1}{1 + e^{-z}}$. The decision boundary is where $\sigma(z) = 0.5$. Setting $\sigma(z) = 0.5$ gives $\frac{1}{1 + e^{-z}} = 0.5$, which implies $1 = 0.5(1 + e^{-z})$, so $2 = 1 + e^{-z}$, and $e^{-z} = 1$. This occurs when $-z = 0$, or $z = 0$. Therefore, the decision boundary is defined by $z = 0$. The linear component of the model is given by $z = w_1x_1 + w_2x_2 + b$. In this problem, $z = 2x_1 - x_2 + b$. Since the decision boundary is.

13 **[Intermediate]** • **Target: Binary Cross-Entropy Loss**

A spam classification model identifies $TP = 120$ genuine spam emails and $FP = 40$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

Final Answer: 75.0%. Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{120}{120 + 40} = \frac{120}{160} = 75.0\%$.

14 **[Intermediate]** • **Target: The Confusion Matrix and Core Metrics**

A spam classification model identifies $TP = 150$ genuine spam emails and $FP = 50$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

Final Answer: 75.0%. Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{150}{150 + 50} = \frac{150}{200} = 75.0\%$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: The F_1-Score, Thresholding and ROC-AUC • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A spam classification model identifies $TP = 180$ genuine spam emails and $FP = 60$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

Final Answer: 75.0%. Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{180}{180 + 60} = \frac{180}{240} = 75.0\%$.

16 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

A spam classification model identifies $TP = 90$ genuine spam emails and $FP = 30$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

Final Answer: 75.0%. Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{90}{90 + 30} = \frac{90}{120} = 75.0\%$.

17 [Challenging] • Target: The Odds Ratio and Logit Formulation • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

A spam classification model identifies $TP = 30$ genuine spam emails and $FP = 10$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

Final Answer: 75.0%. Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{30}{30 + 10} = \frac{30}{40} = 75.0\%$.

18 [Challenging] • Target: Binary Cross-Entropy Loss • Cross-Lesson: 3.01 Gradient Descent and Optimization

A spam classification model identifies $TP = 60$ genuine spam emails and $FP = 20$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

Final Answer: 75.0%. Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{60}{60 + 20} = \frac{60}{80} = 75.0\%$.

19 [Challenging] • Target: The Confusion Matrix and Core Metrics • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

The scatterplot below displays the relationship between the number of hours students spent studying for a test and their scores on that test. The equation of the least-squares regression line is approximately $\hat{y} = 55.2 + 4.8x$, where x is the number of hours spent studying and $\hat{y}$ is the predicted test score.

Final Answer: 4.8. The slope of the least-squares regression line represents the average change in the response variable (test score) for a one-unit increase in the explanatory variable (hours studying). In the given equation $\hat{y} = 55.2 + 4.8x$, the coefficient of x is 4.8. Therefore, for each additional hour a student studies, their predicted test score increases by an average of 4.8 points.

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20 [Challenging] • Target: The F_1-Score, Thresholding and ROC-AUC • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A social scientist is investigating the relationship between the number of hours spent studying per week (X) and the final exam score (Y) for a group of $n = 100$ students. The least-squares regression line is given by $\hat{y} = 55.2 + 8.5x$. The standard error of the slope is $SE_b = 1.2$. The researcher wants to construct a 95% confidence interval for the true slope of the regression line. What is the value of the t-score used for this confidence interval? Show your complete algebraic work.

Final Answer: 1.984. The problem asks for the t-score used to construct a 95% confidence interval for the true slope of a regression line. The degrees of freedom for a simple linear regression are $df = n - 2$, where n is the number of data points. In this case, $n = 100$, so $df = 100 - 2 = 98$. We need to find the t-score such that 95% of the area under the t-distribution curve with 98 degrees of freedom is between $-t$ and t. This means that 2.5% of the area is in each tail ($\alpha/2 = 0.025$). L.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows \(TP = 45 \), \(TN = 40 \), \(FP = 10 \), and \(FN = 10 \). Compute the overall classification accuracy.

a. **\(81.0\% \) ✓**

b. \(81.8\% \)

c. \(52.4\% \)

d. \(13 \)

Accuracy is the ratio of correct predictions to total samples: \(\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{45 + 40}{105} = \frac{85}{105} = 81.0\% \).

22 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows \(TP = 50 \), \(TN = 40 \), \(FP = 10 \), and \(FN = 10 \). Compute the overall classification accuracy.

a. **\(81.8\% \) ✓**

b. \(83.3\% \)

c. \(54.5\% \)

d. \(13 \)

Accuracy is the ratio of correct predictions to total samples: \(\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{50 + 40}{110} = \frac{90}{110} = 81.8\% \).

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23 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 55$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

a. **82.6% ✓**

b. 84.6%

c. 56.5%

d. 13

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{55 + 40}{115} = \frac{95}{115} = 82.6\%$.

24 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 60$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

a. **83.3% ✓**

b. 85.7%

c. 58.3%

d. 13

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{60 + 40}{120} = \frac{100}{120} = 83.3\%$.

25 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 65$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

a. **84.0% ✓**

b. 86.7%

c. 60.0%

d. 13

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{65 + 40}{125} = \frac{105}{125} = 84.0\%$.

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26 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

a. **84.6% ✓**

b. 87.5%

c. 61.5%

d. 13

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{70 + 40}{130} = \frac{110}{130} = 84.6\%$.

27 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

A spam classification model identifies $TP = 30$ genuine spam emails and $FP = 10$ legitimate emails falsely flagged as spam. What is the precision of this model?

a. **75.0% ✓**

b. 60.0%

c. 25.0%

d. 85.7%

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{30}{30 + 10} = \frac{30}{40} = 75.0\%$.

28 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

A spam classification model identifies $TP = 60$ genuine spam emails and $FP = 20$ legitimate emails falsely flagged as spam. What is the precision of this model?

a. **75.0% ✓**

b. 60.0%

c. 25.0%

d. 85.7%

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{60}{60 + 20} = \frac{60}{80} = 75.0\%$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: The Odds Ratio and Logit Formulation

If a classifier predicts probability $p = 0.80$ for disease occurrence, what are the odds of having the disease?

a. 4.0 ✓

b. 0.25

c. 0.80

d. 5.0

Odds $= \frac{p}{1-p} = \frac{0.80}{0.20} = 4.0$ (or 4 to 1).

30 [Intermediate] • Target: The Odds Ratio and Logit Formulation

Compute the log-odds (logit) value when probability $p = 0.50$.

a. 0.0 ✓

b. 1.0

c. 0.5

d. -1.0

$\text{logit}(0.5) = \ln\left(\frac{0.5}{0.5}\right) = \ln(1) = 0.0$.

31 [Intermediate] • Target: The Odds Ratio and Logit Formulation

If a logistic model has equation $\ln\left(\frac{p}{1-p}\right) = 2.0$, find the predicted probability p.

a. $\frac{e^2}{1+e^2} \approx 0.88$ ✓

b. $\frac{1}{2} = 0.50$

c. $\frac{2}{3} \approx 0.67$

d. 0.95

Inverting logit: $p = \sigma(2.0) = \frac{e^2}{1+e^2} = \frac{7.389}{8.389} \approx 0.88$.

32 [Intermediate] • Target: Binary Cross-Entropy Loss

Calculate the binary cross-entropy loss $L = -[y \ln \hat{y} + (1-y)\ln(1-\hat{y})]$ for a sample where true label $y = 1$ and predicted probability $\hat{y} = 0.90$.

a. $-\ln(0.90) \approx 0.105$ ✓

b. $-\ln(0.10) \approx 2.30$

c. 0.90

d. 0.0

When $y = 1$: $L = -1\ln(0.90) - 0 = -\ln(0.90) \approx 0.105$.

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33 [Intermediate] • Target: Binary Cross-Entropy Loss

What is the derivative of the binary cross-entropy loss with respect to the pre-activation logit z, where $\hat{y} = \sigma(z)$?

a. $\hat{y} - y$ ✓

b. $\hat{y}(1 - \hat{y})$

c. $y - \hat{y}$

d. $\frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$

Remarkably clean: $\frac{\partial L}{\partial z} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} = -\left(\frac{y}{\hat{y}} - \frac{1 - y}{1 - \hat{y}}\right)\hat{y}(1 - \hat{y}) = \hat{y} - y$.

34 [Intermediate] • Target: Binary Cross-Entropy Loss

What happens to the cross-entropy loss as $\hat{y} \to 0$ when true label is $y = 1$?

a. **It approaches $+\infty$, heavily penalizing confident wrong predictions** ✓

b. It approaches zero

c. It oscillates around 1

d. It equals the pre-activation logit

$L = -\ln(\hat{y})$. As $\hat{y} \to 0$, $-\ln(\hat{y}) \to +\infty$.

35 [Intermediate] • Target: The Confusion Matrix and Core Metrics

A diagnostic model reports $TP = 70, TN = 20, FP = 5, FN = 5$. Compute the Accuracy.

a. 90% ✓

b. 85%

c. 93.3%

d. 75%

Accuracy $= \frac{TP + TN}{TP + TN + FP + FN} = \frac{70 + 20}{100} = 90\%$.

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36 [Intermediate] • Target: The Confusion Matrix and Core Metrics

Using the same counts ($TP = 70, FP = 5, FN = 5$), calculate the model's Precision.

a. $\frac{70}{75} \approx 93.3\%$ ✓

b. $\frac{70}{70+5} = 90\%$

c. $\frac{70}{100} = 70\%$

d. $\frac{75}{100} = 75\%$

Precision $= \frac{TP}{TP+FP} = \frac{70}{75} \approx 93.3\%$.

37 [Intermediate] • Target: The Confusion Matrix and Core Metrics

Calculate the Recall (Sensitivity) for the model above.

a. $\frac{70}{75} \approx 93.3\%$ ✓

b. $\frac{70}{90} \approx 77.8\%$

c. $\frac{5}{75} \approx 6.7\%$

d. 85%

Recall $= \frac{TP}{TP+FN} = \frac{70}{70+5} = \frac{70}{75} \approx 93.3\%$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The F_1-Score, Thresholding and ROC-AUC • Cross-Lesson: 3.01 Gradient Descent and Optimization

Compute the F_1-score for a model having Precision $= 0.80$ and Recall $= 0.60$.

a. $\frac{2(0.8)(0.6)}{0.8+0.6} \approx 0.686$ ✓

b. 0.700

c. 0.480

d. 0.750

F_1 is the harmonic mean: $2\frac{P \cdot R}{P+R} = \frac{2(0.48)}{1.40} = \frac{0.96}{1.40} \approx 0.686$.

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39 [Challenging] • Target: The F_1-Score, Thresholding and ROC-AUC • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

If you lower the classification probability threshold from 0.5 to 0.2, what is the typical effect on Precision and Recall?

a. Recall increases (fewer false negatives), while Precision typically decreases (more false alarms) ✓

b. Precision increases while Recall decreases

c. Both metrics increase simultaneously

d. Neither metric changes

A lower threshold flags more samples positive: catches more real cases (high Recall), but flags more false alarms (lower Precision).

40 [Challenging] • Target: The F_1-Score, Thresholding and ROC-AUC • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

What does an Area Under the ROC Curve (ROC-AUC) of 0.50 indicate about a classifier?

a. The model has no discriminative ability (equivalent to random guessing) ✓

b. The model is a perfect classifier

c. The model achieves 50% precision at all thresholds

d. The model is inverted

The diagonal line from $(0, 0)$ to $(1, 1)$ represents random guessing and has area under the curve $= 0.50$.

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41 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Consider a simple logistic regression model where the decision boundary is defined by the equation $-1.5x_1 + 2x_2 + 5 = 0$. For a data point (x_1, x_2), if $-1.5x_1 + 2x_2 + 5 < 0$, it is classified into Class 0. Otherwise, it is classified into Class 1. Which of the following points lies on the decision boundary?

a. $(-1, 1)$

b. $(2, -2)$

c. $(0, -2.5)$ ✓

d. $(3, 0.5)$

The decision boundary is defined by the equation $-1.5x_1 + 2x_2 + 5 = 0$. To find which point lies on the boundary, we substitute the coordinates of each point into the equation. For point $(-1, 1)$: $-1.5(-1) + 2(1) + 5 = 1.5 + 2 + 5 = 8.5$, which is not 0. For point $(2, -2)$: $-1.5(2) + 2(-2) + 5 = -3 - 4 + 5 = -2$, which is not 0. For point $(0, -2.5)$: $-1.5(0) + 2(-2.5) + 5 = 0 - 5 + 5 = 0$. This point lies on the boundary. For point $(3, 0.5)$: $-1.5(3) + 2(0.5) + 5 = -4.5 + 1 + 5 = 1.5$, which is not 0. Th.

42 [Challenging] • Target: The Odds Ratio and Logit Formulation • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A researcher is analyzing the relationship between the number of hours a student spends on extracurricular activities per week and their final exam score in a challenging STEM course. The researcher collects data from 100 students and fits a linear regression model. The model is given by $\text{Exam Score} = 65 + 2.5 \times \text{Hours} - 0.3 \times \text{Hours}^2$, where Exam Score is the final exam score and Hours is the number of hours spent on extracurricular activities per week. The researcher believes that for a certain range of hours, there might be an optimal number of extracurricular hours that maximizes the exam score, and beyond that, the score might decrease due to time constraints. \\ \\ Consider the function representing the exam score as a function of hours: $f(h) = 65 + 2.5h - 0.3h^2$. To find the maximum exam score and the corresponding hours, calculus can be used. \\ Which of the following is the approximate number of extracurricular hours per week that would maximize the student's final exam score, assuming the model holds for $h \geq 0$?

a. **4.17 hours** ✓

b. 2.5 hours

c. 5.0 hours

d. 8.33 hours

The exam score is modeled by the quadratic function $f(h) = 65 + 2.5h - 0.3h^2$. This is a downward-opening parabola, so its maximum occurs at the vertex. To find the vertex, we can use calculus and find the critical points by taking the derivative with respect to h and setting it to zero: \\ $f'(h) = \frac{d}{dh}(65 + 2.5h - 0.3h^2) = 2.5 - 0.6h$. \\ Setting the derivative to zero to find the critical point: \\ x. \\ To confirm this is a maximum, we can check the second derivative: $f''(h) = -0.6$. Since the s.

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43 [Challenging] • Target: Binary Cross-Entropy Loss • Cross-Lesson: 1.01 Vectors and Embeddings in AI

The scatterplot below displays the relationship between the number of hours (h) students spent studying for a math test and their scores (s) on that test. The equation of the least-squares regression line for this data is given by $s = 1.2h + 78.5$. If a student studied for 5 hours, what is their predicted score based on this model?

a. 91.5

b. **84.5** ✓

c. 81.5

d. 90.5

The equation of the least-squares regression line is given by $s = 1.2h + 78.5$, where s represents the predicted score and h represents the number of hours spent studying. To find the predicted score for a student who studied for 5 hours, we substitute $h = 5$ into the equation: $s = 1.2(5) + 78.5$ $s = 6.0 + 78.5$ $s = 84.5$ Therefore, the predicted score for a student who studied for 5 hours is 84.5.

44 [Challenging] • Target: The Confusion Matrix and Core Metrics • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider the following quadratic loss function for a regression problem: $L(y, \hat{y}) = (y - \hat{y})^2$. If the predicted value $\hat{y} = 5$ and the true value $y = 3$, what is the value of the loss function?

a. 4 ✓

b. -2

c. 9

d. 2

The quadratic loss function is defined as $L(y, \hat{y}) = (y - \hat{y})^2$. Given the true value $y = 3$ and the predicted value $\hat{y} = 5$, we substitute these values into the loss function: $L(3, 5) = (3 - 5)^2$ $L(3, 5) = (-2)^2$ $L(3, 5) = 4$ Therefore, the value of the loss function is 4.

45 [Challenging] • Target: The F_1-Score, Thresholding and ROC-AUC • Cross-Lesson: 3.01 Gradient Descent and Optimization

The following diagram shows a scatterplot comparing the number of hours studied per week for a math course and the final exam score for a sample of students. What is the approximate y-intercept of the line of best fit?

a. 50

b. **60** ✓

c. 70

d. 80

The y-intercept of the line of best fit represents the predicted score when the number of hours studied is 0. Looking at the scatterplot, if we were to extend the apparent trend of the data back to the y-axis (where hours studied is 0), the line of best fit would cross the y-axis at approximately 60.

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